Triangulations mathematical overview

Setting. A strictly convex polygon has $n \geq 3$ fixed, labeled corners. A triangulation fills it using noncrossing diagonals between original corners only. No new boundary or interior vertices are allowed. Rotating the paper does not identify different labeled fillings.

Counts and local moves. Every triangulation has $n-2$ triangles and $n-3$ diagonals. There are C_{n-2} labeled triangulations, where $C_m = \frac{1}{m+1}\binom{2m}{m}$; the small counts are $1, 2, 5, 14$ for $n = 3, 4, 5, 6$. Each diagonal borders two triangles forming a convex quadrilateral. Replacing it by that quadrilateral's other diagonal is a flip. Thus each triangulation has exactly $n-3$ flip neighbors.

Global structure. The pentagon's five triangulations form a five-cycle under flips, so an odd return route is possible. All triangulations of a fixed convex polygon are connected by flips. If d_v of a triangulation's diagonals already meet corner v, its exact distance to the fan at v is $n-3-d_v$: each flip adds at most one desired diagonal, and a sequence attaining that bound exists. Counting missing target diagonals gives only a lower bound for an arbitrary nonfan target.

Across the bands. K–1 constructs valid fillings, compares counts and makes one-line changes. Grades 2–3 organizes complete small collections and draws a graph whose vertices are whole triangulations. Grades 4–5 investigates odd loops, shortest routes to fans and general connectivity. The Catalan formula is adult orientation, not a required child calculation or a theorem proved in full in this packet.

Discovery and explanation. Let drawings suggest a count or distance conjecture. A complete classification proves a small collection is exhaustive; induction explains the triangle count; a lower bound plus a matching flip route proves a shortest distance. A route found experimentally need not be shortest. The convexity and fixed-vertex assumptions matter. Detailed proofs and solutions follow the preparation page.

Polygon triangulations facilitator guide

This guide, F14-FAC-v5, keys the F14-K-v5, F14-23-v5, and F14-45-v5 packets of the November 6, 2026 kid-fit revision, with 7, 8, and 7 pages; K–1 page 7 is unnumbered extra workspace. The middle packet has seven problems; page 8 is explicitly labeled Problem 7 (continued) and provides extra hexagons for the same collection. There are 20 distinct numbered problems across the three bands. "Week 14" is an unscheduled library label. This guide and these activities are draft and unpiloted.

Mathematical purpose. Children construct complete triangle fillings, distinguish fixed-label configurations, find complete small collections, and investigate how one local change connects them. The pentagon admits a five-flip return loop. Fan targets have a useful exact distance rule. These conclusions should emerge after drawing and comparing.

Prerequisites and entry points

- K–1: point to original corners, draw or direct a straight line, recognize triangles, and compare two pictures. Adult reads; children draw on large printed boards, with no tracing paper. No reading, angle measurement, or multiplication is required.
- Grades 2–3: preserve labels while comparing pictures, organize a collection, and count changes. The hexagon collection is a substantial optional endpoint.
- Grades 4–5: track several configurations and compare a construction with a lower bound. General reasoning may be spoken and picture based. No Catalan formula, algebra, or polytope vocabulary is assumed.

Supplies and preparation for eleven children. Prepare eleven pencils, erasers and short straightedges, about 20 tracing sheets cut into quarters ($4\frac{1}{4} \times 5\frac{1}{2}$ in, which covers the largest board, 3.8×4.4 in), and at least 30 blank Letter sheets beyond the packets. K–1 children draw directly on their large printed boards. The seven older children use a tracing for each find, so plan on about ten quarters each, with a few spare at the K–1 table for Problem 5. These are proposed quantities, not an inventory claim. Print single-sided at 100%. The large working polygons are roughly 2–4.5 inches across. The student polygon boards are regular, with equally scaled axes; compact copies are for recording answers. Use the larger outlines or tracing paper for the main exploration. Tracing sheets preserve printed starts and separate alternative answers. Pattern-block triangles are unsuitable: legal triangles vary in size and shape. Label every traced corner before drawing inside it.

A flexible one hour menu

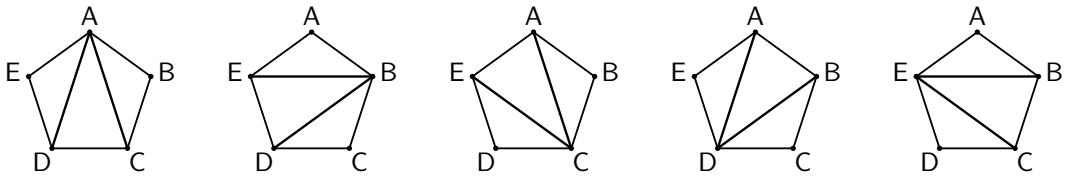
- Minutes 0–5: run around; the children arrive straight from school.
- Minutes 5–9: the whole group launch below; then let children handle straightedges and, from grade 2, tracing paper.
- Minutes 9–28: K–1 P1–2; middle P1–2; upper P1–2. Let children draw and compare before organizing the full collection.
- Minutes 28–50: choose fixed-line completions, one-change neighbors, a pentagon map, or a shortest-route question. Keep middle P7 in reserve if collecting fourteen cases would displace the flips children want to explore.
- Minutes 50–60: share a complete collection, an odd return route, or a shortest-route certificate. Stopping with one convincing example is fine.

Whole group launch. On a convex quadrilateral, draw one diagonal between two printed corners. Ask, "Is the whole shape triangles now?" Trace a copy, erase only that diagonal on the tracing, and draw the other one. Let children make the same change. Then show the referee check every table uses for every change: a partner lays the new drawing on the old one and points to the one line that moved. A drawing with two changed lines, or redrawn from scratch two changes away, does not count as one change. The K–1 table skips the tracing (page 4). Name the operation later, after they have used it. Do not show the five-cycle or the fan rule in advance.

Rules and a complete pentagon collection

Legal fillings. Use only straight segments between the original corners of a strictly convex polygon. No new interior points, extra edge points, moving corners, or crossing interiors. Fill the entire polygon with triangles. The labels stay attached to their corners even if the sheet turns. Two fillings differ exactly when their sets of diagonals differ. Rotations of the paper do not identify labeled fillings.

Reading this key. AC means the diagonal joining A and C, and a list such as AC, AD specifies the entire filling. For K–1 use $1 = A, 2 = B, 3 = C, 4 = D, 5 = E, 6 = F$. Lists never include polygon boundary edges. In the diagrams below, F_A means the fan whose two diagonals meet at A.



F_A AC, AD | F_B BD, BE | F_C AC, CE | F_D AD, BD | F_E BE, CE

Why there are exactly five. A pentagon filling uses two diagonals. If two pentagon diagonals have no common endpoint, their interiors cross, so the two legal diagonals share a corner. Both diagonals from that corner are then forced. Each of the five corners gives one distinct fan. This proves that the displayed list is complete. A second organizing method is to choose the triangle on a fixed boundary edge; the hexagon version appears on page 8.

Why the triangle count stays fixed. An n-corner polygon has $n - 2$ triangles and $n - 3$ diagonals in every triangulation. For induction, a triangle gives the base case. A diagonal divides an n-gon into polygons with p and q corners, where $p + q = n + 2$. The smaller fillings contain $(p - 2) + (q - 2) = n - 2$ triangles. They contain $(p - 3) + (q - 3)$ interior diagonals, plus the dividing diagonal, giving $n - 3$. This argument uses a diagonal in an already complete filling; it does not prescribe how children must build one.

Small counts for this packet

Corners	3	4	5	6	7	8
Triangles in each filling	1	2	3	4	5	6
Diagonals in each filling	0	1	2	3	4	5
Different labeled fillings	1	2	5	14	42	132

The last row counts different whole fillings, not triangles inside one filling. Children need not enumerate the heptagon or octagon in this session.

Common confusions. A line between two neighboring corners is a boundary side, not a diagonal. A line through a new crossing point is illegal even if the pieces look triangular. A graph joining *whole filling pictures* is a new kind of diagram; its joins are not diagonals inside the original polygon. Only the latter are subject to the noncrossing rule.

K–1 solutions for Problems 1 to 4

At the K–1 table. Children draw directly on the large boards, with no tracing. The adult checks each change by finding the one line crossed out and the one line drawn. The rule that every corner of every triangle is a printed dot is what rules out crossing lines: if a child draws a star, point to a crossing that is not a dot. Problem 6 needs only the pentagon, so a table can turn to it straight after Problem 2.

Problem 1

Any two distinct legal fillings of each printed polygon suffice. For the pentagon, use $\{13, 14\}$ and $\{24, 25\}$. For the hexagon, use $\{13, 14, 15\}$ and $\{24, 25, 26\}$. Each pentagon has 3 triangles; each hexagon has 4.

If needed. If a child repeats the same picture, compare the endpoints of every line. Turning the page around does not make a new filling: compare corner numbers, not looks.

Problem 2

There are five fillings, exactly the five fans on page 3 with letters replaced by numbers. Their diagonal sets are $\{13, 14\}$, $\{24, 25\}$, $\{13, 35\}$, $\{14, 24\}$, and $\{25, 35\}$. The page has six boards for five fillings, so one stays empty.

If needed. Let the child keep each discovery on its own board; the packet's last page has spare boards. When ready to check completeness, ask whether the two lines meet at a corner and which corner it is.

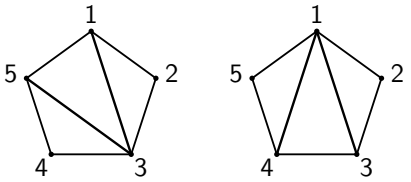
Problem 3

Three triangles are impossible; four are possible. For example, $\{13, 14, 15\}$ gives four. Any diagonal of a hexagon separates a triangle and a pentagon, or two quadrilaterals. Completing those sides gives $1 + 3 = 4$ or $2 + 2 = 4$ triangles. Thus changing the shape of the triangles cannot yield three.

If needed. Ask the child to point to any uncovered region in a three-triangle attempt. For a child seeking a reason that covers every attempt, cover one side of a chosen diagonal and count the two smaller pieces separately.

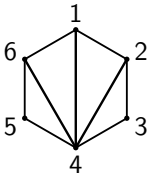
Problem 4

Keeping 13 in the pentagon leaves a quadrilateral with two choices. Keeping 14 in the hexagon leaves two quadrilaterals, each with two independent choices: four completions. Only two pentagon completions exist, so one of the three pre-lined pentagons stays empty, and two of the six pre-lined hexagons stay empty; the spares keep the boards from giving the count away. Here is the full answer collection.

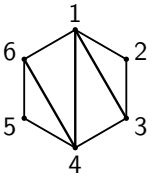


13, 35

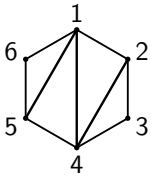
13, 14



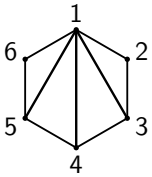
14, 24, 46



13, 14, 46



14, 15, 24



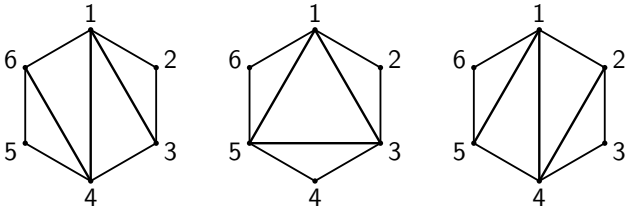
13, 14, 15

If needed. The required diagonal is printed on every board; if a child erases it, redraw it on a spare board from the packet's last page. If one hexagon side is always filled the same way, ask whether its quadrilateral has another filling. Combining each left choice with each right choice proves completeness.

K–1 solutions for Problems 5 and 6

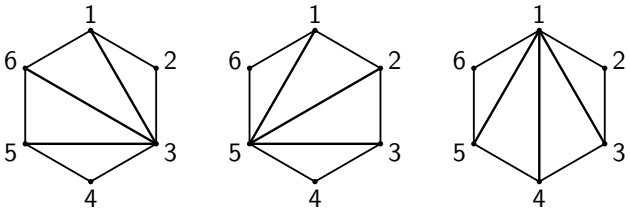
Problem 5

Each printed hexagon has exactly three one-change results, one for each of its three interior diagonals. The original filling itself is not a new result. Each small copy prints its start's lines in light gray: the child crosses out the gray line that goes and draws the new one, so every copy differs from its start by one line. Show this once at the table. Four copies for three results keep the count open. **Handling the small copies.** The copies are about 1.3 in across, and gray lines cannot be erased, so "erase" means cross out: the child slashes the gray line that goes and rules the new one, and a wobbly line that clearly joins the right two dots counts. If small hands cannot manage, the adult rules the line the child points to, the child works on a quarter tracing sheet over the large board, or the adult uses a whiteboard copy, where a line really can be erased. The child still chooses every change. **First start** $\{13, 14, 15\}$ has these three neighbors:



13, 14, 46 13, 15, 35 14, 15, 24

Second start $\{13, 15, 35\}$ has these three neighbors:



13, 35, 36 15, 25, 35 13, 14, 15

Why the list is complete. Removing one interior diagonal merges exactly its two adjacent triangles into a convex quadrilateral. That quadrilateral has exactly one other diagonal. Everything outside it is unchanged. Thus each of the three original diagonals supplies exactly one neighbor, and there are no further legal one-line replacements.

If needed. Trace the two triangles touching the line a child wants to erase. Ask for the boundary of their combined piece before asking which other line fills it. Do not erase a second line.

Problem 6

Yes. Name a pentagon filling by the corner where its two diagonals meet. The route $1, 3, 5, 2, 4, 1$ visits every other filling once and returns in five changes. The route in the opposite direction works too. Its complete diagonal sets are

$$\{13, 14\} \to \{13, 35\} \to \{25, 35\} \to \{24, 25\} \to \{14, 24\} \to \{13, 14\}.$$

Each arrow replaces exactly one line. The pentagon map on page 6 shows why all five fillings have been visited. The ring's top board is the fan at 1; four blank boards take four changes, and the last arrow home is the fifth. Only the first change is a choice; each later one is forced, since the other change goes back.

If needed. Lay the five Problem 2 boards beside the ring. Ask which of them can follow the current picture after one change. A child who reaches a filling already on the ring has gone back; look for the other change.

Useful stopping point. Children who build and explain this five-step loop have encountered a genuine odd cycle. They do not need the word "parity" or the older packet's distance formula.

Grades 2–3 solutions for Problems 1 to 5

Problem 1

Any two distinct legal fillings suffice. Hexagon examples: $\{AC, AD, AE\}$ and $\{BD, BE, BF\}$. Heptagon examples: $\{AC, AD, AE, AF\}$ and $\{BD, BE, BF, BG\}$. The counts cannot differ: every hexagon filling has 4 triangles and every heptagon filling has 5. Use page 3's splitting argument if children seek an explanation.

If needed. Ask whether a thin or large triangle should still count as one. The task uses all triangles, regardless of size or shape.

Problem 2

Exactly five pentagon fillings, displayed on page 3. The shared-corner argument there proves completeness.

If needed. Sort a child's collection by the corner shared by its two diagonals. Keep corner names fixed so that all five fans remain distinct.

Problem 3

First start $\{AC, AD, AE\}$ has neighbors $\{AC, AD, DF\}$, $\{AC, AE, CE\}$, and $\{AD, AE, BD\}$. Second start $\{AC, AE, CE\}$ has neighbors $\{AC, CE, CF\}$, $\{AE, BE, CE\}$, and $\{AC, AD, AE\}$. The six diagrams on page 5 show all results after replacing number labels by letters.

If needed. Choose one original diagonal; identify its two incident triangles. The other diagonal of their quadrilateral is the only legal replacement. Each small copy prints its start in light gray: cross out the gray line that goes and draw the new one.

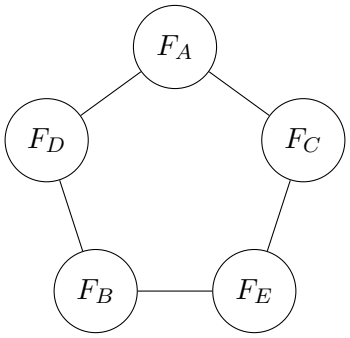
Problem 4

An odd return route is $F_A, F_C, F_E, F_B, F_D, F_A$, with five flips. The complete sets are the K–1 P6 route with letters replacing numbers. No shorter odd return is possible: the pentagon's flip map (Problem 5) has no loops or triangles. An immediate out-and-back takes two flips and does not meet the odd condition. Children find this route before drawing the map; Problem 5 then shows it as the map's cycle.

If needed. Ask children to count *moves*, not pictures, along their route. Six listed pictures represent five flips when the first is repeated at the end.

Problem 5

The complete pentagon map has exactly five joins: F_AF_C, F_CF_E, F_EF_B, F_BF_D, F_DF_A. There are no other joins. The revised student layout follows this cycle around the page.

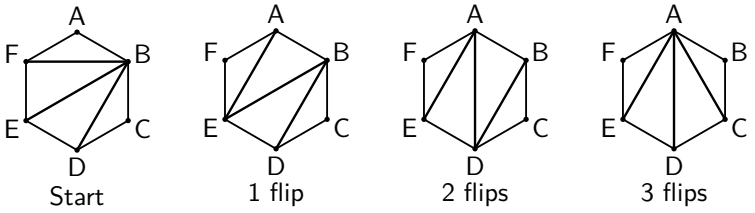


Each adjacent pair keeps one diagonal and replaces the other. Each of the five fillings has only two interior diagonals, hence exactly two possible flips; the displayed joins account for both and prove completeness.

Grades 2–3 Problem 6 and shortest routes

Problem 6

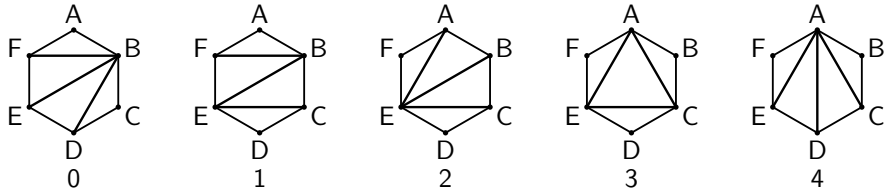
The printed start is the B fan $\{BD, BE, BF\}$ and the target is the A fan $\{AC, AD, AE\}$. The minimum is three flips. Here is one shortest route, read left to right:



Step	Diagonals in the filling	Change just made
0	BD, BE, BF	Start
1	AE, BD, BE	BF → AE
2	AD, AE, BD	BE → AD
3	AC, AD, AE	BD → AC

Why three is best. None of the three target diagonals is present initially. One flip adds only one diagonal, so at least three flips are necessary. The displayed route achieves three. Counting missing target diagonals is a lower bound for arbitrary targets; it is always attainable for a fan target (page 11).

A different simple route has four flips. The following five pictures are all different, so this meets the revised requirement to avoid visiting a filling twice.



Step	Diagonals in the filling	Change just made
0	BD, BE, BF	Start
1	BE, BF, CE	BD → CE
2	AE, BE, CE	BF → AE
3	AC, AE, CE	BE → AC
4	AC, AD, AE	CE → AD

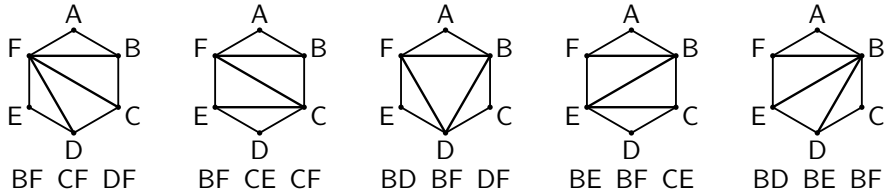
If needed. For the shortest route, ask which target lines are missing and how many can appear in one move. For the second route, try a different first move. Undoing and redoing a move does not satisfy the no-repeated-fillings instruction. Routes are written on the ruled lines as lists of flips, such as "erase BF, draw AE" (Problem 5's example); the working hexagon is for trying moves.

Check the actual move. Two diagonal lists of length three are flip neighbors when they share exactly two diagonals. Merely rotating a page is not a move; nor may the child replace two diagonals at once.

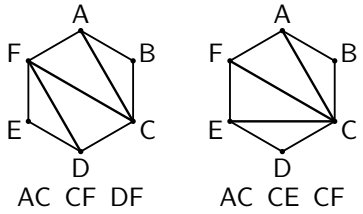
Grades 2–3 Problem 7 and all fourteen fillings

Problem 7

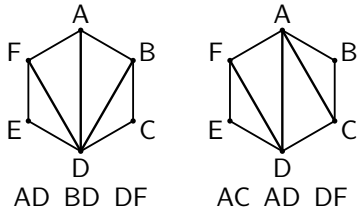
Student page 8 is labeled "Problem 7 (continued)". It supplies recording space for the same problem, not a second task. The complete hexagon collection has 14 fillings. The catalog below is organized by the triangle touching boundary edge AF. Its third corner is B, C, D, or E. **Third corner B: five fillings.** Diagonal BF cuts off triangle ABF, leaving a pentagon with five choices.



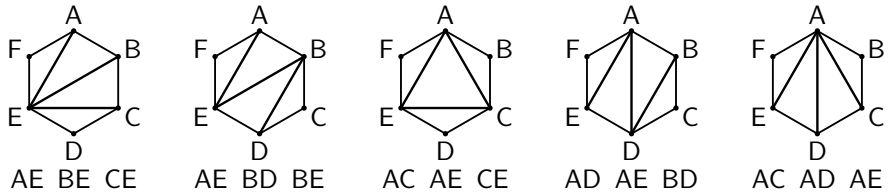
Third corner C: two fillings. Triangle ACF leaves triangle ABC and quadrilateral CDEF.



Third corner D: two fillings. Triangle ADF leaves quadrilateral ABCD and triangle DEF.



Third corner E: five fillings. Triangle AEF leaves pentagon ABCDE.



Why no omissions or repeats. A filling has exactly one triangle touching AF. Its third corner assigns it to exactly one group. The remaining polygon pieces can be filled independently; their complete smaller lists give $5 + 2 + 2 + 5 = 14$. No filling belongs to two groups.

If needed. Let children collect freely first. When a collection is hard to audit, ask which triangle touches AF in each picture. The student pages offer 17 recording outlines; the number of blank spaces is not the answer.

Grades 4–5 solutions for Problems 1 to 4

Problem 1

The five pentagon fans on page 3 are the complete answer. The shared-corner proof or the root-edge classification proves no filling is missing.

Problem 2

The two printed hexagon starts have the same six neighbors keyed in middle P3 and pictured in K–1 P5. Each start has exactly three possible flips. The two-triangle quadrilateral argument on page 5 proves completeness.

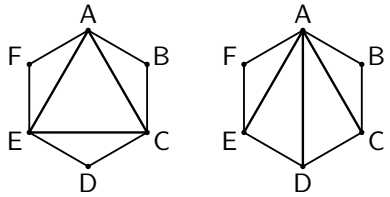
If needed. A different-looking triangulation need not be one move away. Check that exactly one old diagonal disappeared and exactly one new one appeared. Each small copy prints its start in light gray: cross out the gray line that goes and draw the new one.

Problem 3

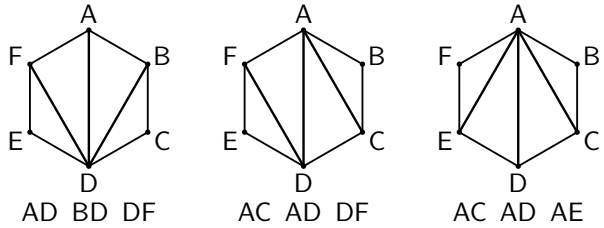
The complete map is the five-cycle on page 6. The five-flip loop $F_A, F_C, F_E, F_B, F_D, F_A$ is an odd return. Its reverse is also valid. The graph is not bipartite: alternating two colors around the cycle makes the final edge join two same-colored vertices. **Fidelity warning.** A parity invariant from lozenge tiling flips does not transfer here. The objects and moves are different; a polygon five-cycle is a direct counterexample to an even-return claim.

Problem 4

The three starts, left to right, are $\{BD, BE, BF\}$, $\{AC, AE, CE\}$, and $\{AD, BD, DF\}$. The fourth filling, where the arrows point, is the A fan $\{AC, AD, AE\}$. Their minimum distances are 3, 1, and 2. **Left start.** Use the three-flip route on page 7: replace BF by AE, BE by AD, then BD by AC. It is shortest because all three target diagonals were missing. **Middle start.** Replace CE by AD. Exactly one target diagonal was missing.



Right start. Replace BD by AC, then DF by AE. Exactly two target diagonals were missing.

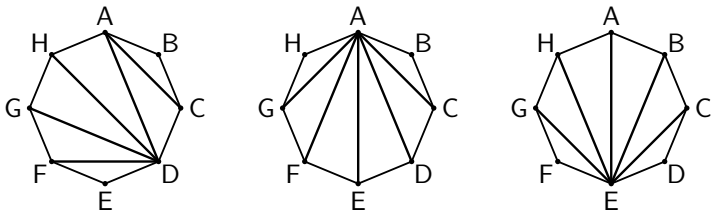


If needed. Have a child mark which A-fan diagonals are already present, without erasing them. Ask how much that count can rise in one flip. A successful short route alone is not a proof that it is shortest. Each route is written as a list of flips on the ruled lines beside the blank hexagon, which is for trying moves.

Grades 4–5 Problem 5 and two octagon fans

Problem 5

The printed start is $S = \{AC, AD, DF, DG, DH\}$. A fan at A has five diagonals $\{AC, AD, AE, AF, AG\}$; a fan at E has $\{AE, BE, CE, EG, EH\}$. The minimum distances are 3 and 5, respectively.



Three-flip route to the A fan

Step	Diagonals in the filling	Change just made
0	AC, AD, DF, DG, DH	Start
1	AC, AD, AG, DF, DG	DH → AG
2	AC, AD, AF, AG, DF	DG → AF
3	AC, AD, AE, AF, AG	DF → AE

Five-flip route to the E fan

Step	Diagonals in the filling	Change just made
0	AC, AD, DF, DG, DH	Start
1	AC, AD, DG, DH, EG	DF → EG
2	AC, AD, DH, EG, EH	DG → EH
3	AC, AD, AE, EG, EH	DH → AE
4	AC, AE, CE, EG, EH	AD → CE
5	AE, BE, CE, EG, EH	AC → BE

Optimality certificates. The start already has two of A's five diagonals and none of E's five. Each flip can add at most one target diagonal. Therefore at least $5 - 2 = 3$ and $5 - 0 = 5$ flips are required. Both tables achieve their lower bounds, always adding a target diagonal and never deleting one.

If needed. Ask the child to choose a target corner and point to every diagonal already meeting it. If a proposed flip removes one of those, ask whether a different choice can increase the count instead. The general reason that such a choice exists is on page 11.

Using the tables. Step 0 is the printed start. The change column names the entire local move; all other diagonals stay fixed. Children write each route on the ruled lines as a list of flips, such as "erase DH, draw AG", and try moves on the blank octagon. Nine lines under each fan leave room for the five-flip route and a false start.

Grades 4–5 Problem 6 and the fan rule

Problem 6

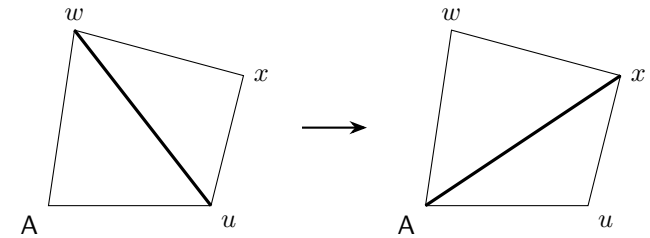
For a convex n-gon, let d_A be the number of *interior diagonals* in the starting filling incident to A. The minimum number of flips to the A fan is

$$\boxed{n-3-d_A}.$$

Do not count the two boundary sides at A. For a triangle, $n-3=d_A=0$, so the formula includes the zero-move case. The page gives a blank hexagon, heptagon and octagon for testing a rule.

Lower bound. Every filling has $n-3$ diagonals. An A fan has all of them incident to A. One flip adds at most one new A diagonal, so at least $n-3-d_A$ flips are necessary.

Why that bound can always be attained. Look at the union of all triangles touching A. If it is not the whole polygon, part of its boundary is an interior diagonal uw, with triangle Auw on one side and a triangle uwx on the other. Their union is a convex quadrilateral. Flip uw to Ax. The old edge is not incident to A, and the new one is. The union of triangles at A grows, and the number of A diagonals rises by exactly one. Repeat until every diagonal meets A. There are exactly $n-3-d_A$ increases.



Before: uw separates two triangles After: one more diagonal at A

The picture is a local part of a larger polygon. Outside diagonals stay fixed. Strict convexity guarantees the other diagonal lies inside this quadrilateral.

If needed. First shade every triangle touching A on one concrete drawing. Ask where the shaded region could expand by changing one line. Once the child can do it twice, ask what would have to be true for the same move to be available again.

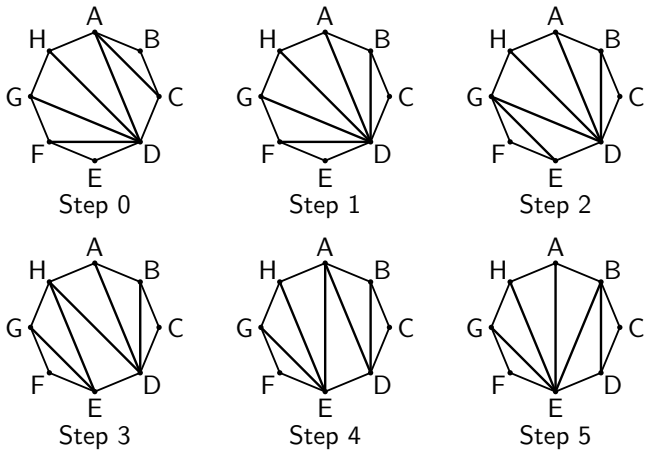
What constitutes an explanation. "I always add an A line" is a useful strategy but incomplete unless the child explains why such a flip exists whenever some A lines are missing. The boundary-of-the-shaded-region argument supplies that missing step. The lower bound and construction have different jobs, and both are needed for an exact minimum.

Why convexity matters. In a nonconvex polygon, the union of two triangles need not be a convex quadrilateral; its other diagonal may leave the polygon. The guide's always-available flip argument is only for the strictly convex printed polygons.

Grades 4–5 Problem 7 and connectivity

Problem 7

The printed endpoints are $S = \{AC, AD, DF, DG, DH\}$ and $T = \{AE, BD, BE, EG, EH\}$. Here is a five-flip route.



Step	Diagonals in the filling	Change just made
0	AC, AD, DF, DG, DH	Start
1	AD, BD, DF, DG, DH	AC $\rightarrow$ BD
2	AD, BD, DG, DH, EG	DF $\rightarrow$ EG
3	AD, BD, DH, EG, EH	DG $\rightarrow$ EH
4	AD, AE, BD, EG, EH	DH $\rightarrow$ AE
5	AE, BD, BE, EG, EH	AD $\rightarrow$ BE

This route is also shortest, although the problem does not ask for that: the endpoints share no diagonal, so all five target diagonals must be introduced and one move adds at most one.

Why any two fillings can be joined. Choose one fixed corner, say A. Page 11's growing-fan procedure turns the first filling into the A fan and the second filling into the same A fan. Reverse the second route and place it after the first. Every reversed flip is legal, since exchanging the two diagonals of a convex quadrilateral is reversible. This gives a route between the original fillings. It need not be shortest.

If needed. If children can reach a fan but cannot connect arbitrary endpoints, ask whether a route can be followed backward. Be explicit that the same target corner must be used for both routes. A route can be recorded on the one working board as a list of changes, such as "erase AC, draw BD" from Problem 3's example, with only the final filling drawn.

An upper bound, not an exact distance. Passing through one fan uses $(n-3-d_A(S)) + (n-3-d_A(T))$ flips. For these endpoints A gives $3+4=7$, while the route above uses 5. This example shows why a connectivity construction should not automatically be called a shortest route.

Extensions sources and verification

Extensions with clear endpoints

- Find the two fillings of a quadrilateral and their flip map. It has two vertices and one edge; every return walk is even. Compare this with the pentagon five-cycle. Odd return behavior depends on the configuration graph.
- Choose boundary edge AF of the hexagon and count by its incident triangle. Then repeat the idea on a heptagon: the five possible third corners give $14 + 5 + 4 + 5 + 14 = 42$ fillings. Children may explain the products using independent polygon pieces; do not require the formula first.
- Generalize that decomposition. If T_n counts n-gon fillings and $T_2 = 1$ represents an empty side of a root triangle, then $T_n = \sum_{k=1}^{n-2} T_{k+1}T_{n-k}$. With $C_m = T_{m+2}$, this is $C_0 = 1$ and $C_{m+1} = \sum_{i=0}^{m} C_i C_{m-i}$. The empty-piece convention is a counting tool, not a claim that a two-sided polygon exists.
- For a nonfan target, the missing-diagonal bound can be strict. From $\{BF, CE, CF\}$ to $\{AD, AE, BD\}$, all three target lines are missing, but the first flip can add only DF, BE, or AC, none of them a target line. At least four flips are needed. Four suffice via $\{BF, CF, DF\}$, then $\{BD, BF, DF\}$, then $\{AD, BD, DF\}$, then the target.

Sources and exact scope

- Richard P. Stanley, *Catalan Numbers*, Tsuda slides (2017), PDF pp. 16–18 and 26–28, for polygon counts and the root-edge decomposition: `https://math.mit.edu/~rstan/transparencies/tsuda.pdf`.
- Vincent Pilaud, *Multitriangulations, pseudotriangulations and some problems of realization of polytopes*, Chapter 1, §1.1, Lemma 1.2 and Proposition 1.4; printed pp. 51–53, PDF pp. 68–71, for triangle counts and the fan connectivity argument: `https://www.ub.edu/comb/vincentpilaud/documents/reports/theseVincentPilaud.pdf`.
- Givental, Nemirovskaya, and Zakharevich, *Math Circle by the Bay* (AMS, 2018), Preface pp. vii–ix. Its concrete manipulatives, student interaction, fallback hints, and repeated attention to explanations informed the adult support here. The exact menu and launch are local suggestions.

These are mathematical and pedagogical background sources, not classroom trials of these worksheets. Specific configurations, full answer catalogs, shortest routes, hints, and diagrams here were made for the final packet. Associahedra, parenthesizations, and generalized triangulations are not student prerequisites. The result assumes fixed labeled corners, straight diagonals, no added vertices, and strict convexity. Counts would change under different equivalences or allowed moves.

Independent checks. Root-edge recursion generated all triangulations for 3 through 8 corners, with counts 1, 2, 5, 14, 42, 132. Breadth-first search checked every printed route example and every fan distance through eight corners. Each listed route was checked for legal successive flips; the alternative middle route has no repeats. The checks were written independently of worksheet builders and draft count files. General proofs, rather than finite testing, support claims for all polygon sizes.

After use. Record which problems children actually attempted, whether labels remained fixed, whether the flip action was clear, which hints helped, and any explanation worth preserving. Do not mark the material piloted until that classroom evidence exists.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

Keep one adult at each of the three tables; adults can rule or record a child-chosen filling. On a return visit, K-1 can keep a line while completing a polygon (P4) or change one line (P5-6); P6 needs only the pentagon and can follow P2 directly. Grades 2-3 can use their pentagon gallery for the odd return and the flip map (P4-5), with the hexagon one-flip and shortest-route problems (P3, P6), leaving the full hexagon collection (P7) for a separate investigation. Grades 4-5 can revisit shortest fan routes (P4-5) and then the general fan rule or connectivity (P6-7). Preserve labeled boards, tracings and written routes between visits.

Current child count

Provide eleven pencils/straightedges for the current 4/4/3 children and about 20 tracing sheets cut into quarters, about ten quarters for each of the seven older children (page 2). Since the November 6, 2026 kid-fit revision (F14-*-v5), K-1 draws on large printed boards with no tracing paper, and the older bands write routes as lists of flips.